A parameterised water balance driven by a frozen foundation model

Soil moisture project — Session 30

2026-08-24

Contents

Design document for critique. Nothing built. No GPU spent.

Session 30, 2026-08-24. Supersedes nothing yet — this is an alternative to §31/§32's per-location processor, not a replacement for it. §32's terrain pipeline is a dependency of both and is unaffected.

The idea in one sentence

> *The network does not predict soil moisture. It predicts **the properties of the ground**. A differentiable water balance predicts soil moisture from those properties and the weather.*

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0. FEASIBILITY

This section is first because it is the part that can kill the design, and because two of its findings changed the architecture rather than merely assessing it.

0.1 What the data actually gives

Measured 2026-08-24 from `csvs/station_splits.csv`:

quantity	value
stations	993 — train 683, val 91, oos 219
station-years	7,164 (mean 7.21, median 6.90, range 3–12)
station-days	2,614,787
location groups	906
Köppen classes / networks	25 / 33
colocated pairs	75, across 84 stations
current baseline	`cls_depth_star_reg`, 16 epochs, 10h13m, 4×H100
current ubRMSE 0–10	0.0539
current overfit	21× — train 0.000099 vs val 0.002177

The label count is abundant; the *effective* sample size is not, and it differs per parameter:

```
the shared dynamics  R(t), E(t), m, Q0       ~2.6 M station-days   abundant
the map  description -> (lambda, alpha, beta)   ~993 locations     THE CONSTRAINT
the within-tile contrast  f, pair loss           75 pairs          thin
```

One station per tile means anything about *between-location* differences is a 993-sample problem, not a 2.6-million-sample one. This is the same wall §27b hit, and it is the honest centre of the feasibility question.

0.2 Finding — trajectory output is a 365× win

Today one sample is one station on one date, so a forward pass yields **one label*. *This design emits a* trajectory*: the water balance runs over the 365-day window the transformer already reads, so every day of that window comes out at no extra cost.

```
labels per forward pass          1        ->   up to 365 x 3 = 1,095
forward passes for full coverage 2.6 M    ->   7,164  (one per station-year)
```

Honest caveat: 7,164 samples at batch 128 is roughly **56 optimiser steps per epoch**. Few steps, far richer gradients per step. The optimisation dynamics are unknown and this is a genuine open question — mitigate with random window start dates as augmentation, which multiplies the sample count without new data.

0.3 Finding — compute is a non-issue

The classical TOPMODEL result that makes this cheap: the catchment deficit evolves on the **TWI histogram (~20 bins), not on the grid**. The saturated fraction is an integral over the TWI distribution, so no per-cell loop is needed for the far-field term at all.

```
TRAINING    365 steps x (20 TWI bins + 2 supervised cells)  ~ 8 k scalar ops/sample
INFERENCE   365 steps x 4,900 cells                         ~ 1.8 M scalar ops
AUTOGRAD    365 x B x cells x 2 stores x 4 bytes            < 1 MB at B=128
OVERHEAD    ~5-10 % against a 50 M-param transformer over 1,035 tokens
```

The deleted U-Net decoder — four upsampling stages at 512/256/128/64 plus three FiLM'd skips, ~3.2 M activations per sample per depth — more than pays for the loop. The only real engineering risk is that the loop is **365 sequential kernel launches**; it is launch-bound, not memory-bound. Measure it in the smoke run and chunk or `scan` it if it bites.

0.4 Finding — the 21× overfit is structurally capped

This is the strongest feasibility argument, and it is structural rather than a penalty.

```
TODAY   station identity  ->  decoder  ->  arbitrary value at any date
        measured overfit 21x;  §19.5 proved regularisation cannot touch it;
        the diagnosis was station memorisation

HERE    station identity  ->  lambda, alpha, beta, p, delta  ->  water balance
                              five bounded scalars
```

To memorise a 2,600-day series you would have to route it through five numbers whose ranges are fixed by sigmoid and softplus. That is not discouraged — it is impossible. The model's capacity to fit a station is bounded by the expressive power of a two-store bucket, which is exactly the point.

0.5 Finding — a leak that must be closed, or this is worse than today

If `R(t)` and `E(t)` are read off `ctx[:, era5:]`, those day-tokens have already attended to the station's own 196 spatial tokens. That supplies 365 × 2 station-specific, date-specific numbers — **reopening the memorisation channel and defeating §0.4 entirely**.

```
REQUIRED CHANGE

  R(t), E(t) = MLP( [ raw ERA5 row(t) , g_static ] )        ~5 k params

  g_static = pooled soil / terrain / LULC summary ONLY.
             No per-date content. No station-identifying content. Ever.
```

This costs a small MLP rather than a second transformer, and it is not optional. It was found by the feasibility pass, not by the design.

0.6 Finding — spin-up

```
SURFACE STORE   lambda ~ 0.9 forgets its initial condition in ~30 days
                (0.9^30 = 0.04). A 365-day window is ample.         OK

CATCHMENT S      recession scale m may be months, so 365 days of
                spin-up may not be enough.                          RISK
                Fix: S is ONE scalar. Run it on a 3-year ERA5 window
                at negligible cost, or warm-start from a climatology
                predicted from g. Cheap either way.
```

0.7 The decisive pre-GPU experiment

The entire architecture rests on one untested assumption: **that a location's water-balance parameters are predictable from what the landscape looks like.** That is directly measurable on CPU, before any GPU is spent.

```
STEP 1   Calibrate a three-parameter bucket to each station's OWN daily
         series, forced by that station's own ERA5. 993 independent fits,
         Pool(64), sbatch, minutes.

         Yields:
           (a) fitted lambda-hat, alpha-hat, beta-hat per station
           (b) THE CEILING — how well can the best possible bucket fit at
               all? If the best fit is worse than the current 0.0539 ubRMSE,
               the physics is too crude and the approach caps below the
               baseline. Hard stop, measured in an afternoon.
           (c) per-station equifinality, by profile likelihood

STEP 2   Regress the fitted parameters on tokens + terrain + soil, reusing
         station_mean_probe.py's ladder: fit_block(), GroupKFold on
         location_group_id, RidgeCV as the deliberately weak learner,
         HistGradientBoostingRegressor as the nonlinearity control.

STEP 3   Read the skill.
           > 30 %   the architecture has a foundation. Build it.
           ~ 10 %   marginal. This is where mean SM landed (§27b: 9.8 %).
           ~  0 %   the map does not exist. Do not build it.
```

This is the same shape as §29 Phase A and §32.8's sufficiency gate: a cheap measurement that can kill the arm before the allocation is spent. It is worth more than every other page of this document.

Note the target is **not** the one §27b tested. §27b regressed *mean soil moisture*, which is largely climate and sensor calibration, and got 9.8 % globally and 0–4.2 % within-network. Here the target is a *drainage rate* — a landscape property, which is precisely what §27a's positive controls proved the tokens do encode — and its predictors include TWI, HAND, slope and texture as direct measurements rather than embeddings. Different question, and it may have a different answer. It may also not.

0.8 Data availability

```
ON DISK          labels; ERA5 365-day windows; soil (21,74,74) = 3 depths
                 x 7 properties; S1/S2 rasters at 10 m; TerraMind tokens
IN PROGRESS      TWI / HAND — 353/353 regions fetched, build_twi_hand.py
                 running, MERIT gate (§32.6) not yet run
NEVER MEASURED   per-station Sentinel-1 acquisition counts. Needed to size
                 the assimilation rung (V5) and nothing else.
TO DOWNLOAD      nothing
```

0.9 What would make it infeasible

```
1.  §0.7 Step 1's bucket ceiling worse than 0.0539 ubRMSE     -> dead
2.  §0.7 Step 3's skill ~ 0                                   -> dead
3.  §32.8's sufficiency gate flat both wet and dry            -> V4 dies;
                                                                V1-V3 may not
                                                                beat baseline
4.  56 steps/epoch too few to optimise                        -> fixable by
                                                                window-start
                                                                augmentation
```

Failures 1 and 2 are cheap and come first. Failure 3 prunes one input rather than the architecture, because the per-cell parameters do not depend on terrain alone.

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1. THE DIAGNOSIS

The model reproduces **temporal** variation almost perfectly and **spatial** variation barely at all.

```
within-station temporal SD          0.051   vs observed 0.051
between-station SD                  15-19 % of observed
r(pred level, obs level), CR200-18  -0.175          anti-correlated
§23 norm-std                        0.093-0.524, HIGHEST at the flattest site
```

The cause is verified at `model.py:736-743`. `context` is the masked mean of every valid non-CLS, non-spatial, non-pad token, applied by FiLM as a **uniform** per-channel scale and shift. All time-series signal — ERA5's 365 days, the S2 and S1 histories, SIF, TWSA — reaches the decoder through **one spatially constant vector**.

```
Two stations 500 m apart can be given different MEANS.
They cannot be given different BEHAVIOUR.
```

That is exactly the observed signature: temporal skill perfect, between-station skill absent. The map is not flat — it is wrong.

Three further measurements constrain any fix:

- **§27a** — the tokens carry genuine within-tile spatial structure at 160 m, and it tracks the landscape. Both positive controls passed: the LULC PCA→RGB reproduces the land-cover map, the DEM PCA→RGB traces the drainage. Sink- excluded PCA ratios are high-rank at every depth.
- **§27b** — the same tokens do *not* predict wetness. Own-token within-network skill 0–4.2 %; global 9.8 % against a tabular stack's 23.4 %. The UMAP organises by Köppen class, not by soil moisture.
- **§29.15** — the Landsat LST tile pattern is +0.967 spatially coherent across all twelve months. It is a static landscape field, and *a static field cannot track a dynamic variable*. That is why the pooled correlation was +0.167 and the within-station correlation −0.077: the same fact seen twice.

The design principle follows directly. **Ask each source only for what it has been measured to be good at.**

```
ERA5 weather      the dynamics, shared    temporal SD 0.051 = observed 0.051
imagery/terrain   the PARAMETERS          tracks landscape (§27a) but not
                  not the wetness         wetness (§27b)
soil/climate      the absolute level      23.4 % skill vs the network's ~18 %
tables
Sentinel-1        corrections to state    the only time-varying 10 m carrier
```

---

2. THE FLOWCHART

```
+==============================================================================+
| OFFLINE  precomputed, on disk, never inside the training loop                |
+==============================================================================+

 TerraMind ViT-B (frozen)      §32 regional flow fields      soil rasters
 S2 S1 DEM LULC -> tokens      TWI_k  HAND_k  TWI_mean       OpenLandMap
 (N,196,768) fp16              at FULL catchment scale       -> theta_r, AWC,
        |                              |                        n_e, L
        +--------------------+---------+-----------------------+
                             v
+==============================================================================+
| BLOCK A   DESCRIBE                              existing code, unchanged     |
+==============================================================================+

 seq = [ DEM x4 | LULC x4 | soil x4 | spatial x196 | S2hist | S1hist | ERA5 x365 ]
                                  ^
                                  |  terrain_stem, ADDED, zero-init
                                  |  conv stride-5 over (TWI, HAND, slope, curv)
                        +---------+----------+
                        |  6 x self-attention |
                        +---------+----------+
                                  |
                            ctx (B,1035,768)
                                  |
     +----------------------------+---------------------------+
     v                            v                            v
 ctx[:,12:208]              masked mean                 raw ERA5 row(t)
 196 cells                  g (B,768)                   + g_static
                                                        (NOT from ctx -- §0.5)

+==============================================================================+
| BLOCK B   PARAMETERISE                       four small heads, ~24 k params  |
+==============================================================================+

+-----------------+ +----------------+ +---------------+ +-------------------+
| B1  CELL        | | B2  FORCING    | | B3  READOUT   | | B4  CATCHMENT     |
|                 | |                | |               | |                   |
| gather x5       | | MLP([era5(t),  | | pedotransfer  | | MLP(g -> 2)       |
|  -> 70x70       | |      g_static])| | on soil       | |                   |
| + fine channels | |                | |               | |  m   mm           |
| MLP(768+C -> 5) | |  R(t)  mm/d    | |  theta_r      | |  Q0  mm/d         |
|                 | |  E(t)  mm/d    | |  AWC          | |  f   FIXED        |
|  lambda_k       | |                | |  n_e, L       | |                   |
|  alpha_k        | | SHARED over    | |  NOT LEARNED  | | ONE catchment     |
|  beta_k         | | all cells      | |               | | per tile          |
|  p_k            | |                | | + delta_d     | |                   |
|  K_k            | | (B,365,2)      | |   small, learn| | (B,2)             |
| (B,4900,5)      | |                | | (B,4900,3,2)  | |                   |
+--------+--------+ +-------+--------+ +-------+-------+ +---------+---------+
         +------------------+------------+--------+--------------+
                                          v
+==============================================================================+
```

```
| BLOCK C   SIMULATE                    365 days x 4,900 cells @ 32 m         |
+=============================================================================+

 for t = 1 .. 365:

 +---- FAR FIELD ---- the catchment as ONE state ----------------------------+
 |                                                                           |
 |   S(t)      catchment storage deficit, mm                                 |
 |   D_k(t)  = S(t) + (1/f)*( TWI_mean - TWI_k )        local deficit, mm    |
 |             |moves|        |---- fixed, from §32 ----|                    |
 |   z_k(t)  = D_k(t) / (1000*n_e)                      water table depth, m |
 |   u*_k(t) = exp( -z_k(t) / L_k )                     equilib. saturation  |
 |                                                                           |
 |   D_k <= 0  ==>  SATURATED: u1 = 1, incoming R becomes runoff             |
 +------------------------------------+--------------------------------------+
                                    |  Phi_k = omega * max(0, u*_k - u_k)
                                    |          capillary rise, ONE-SIDED
                                    v
 +---- NEAR FIELD ---- per cell k -------------------------------------------+
 |                                                                           |
 |  u1(t+1) = clip[ lambda_k*u1  + alpha_k*R(t)  - beta_k*E(t)*u1            |
 |                               - p_k*u1       + Phi1_k                     |
 |                  ( + kappa_k*[DIV_W u1](k)   V6 only )   , 0, 1 ]         |
 |                                                                           |
 |  u2(t+1) = clip[ lambda2_k*u2 + p_k*u1 - r_k(t) + Phi2_k , 0, 1 ]         |
 +------------------------------------+--------------------------------------+
                                    |
                      +-------------v--------------+
                      | on Sentinel-1 dates:       |
                      |  sigma_hat = h(u1, rough,  |
                      |                 veg)       |
                      |  u1 <- u1 + K_k*(sigma_obs |
                      |                - sigma_hat)|
                      +-------------+--------------+
                                    |
               dS/dt = Q0*exp(-S/m) - mean_k( r_k )
                        |baseflow out|   |recharge in|
                                    v
                      u1(k,t), u2(k,t)   all cells

+=============================================================================+
| BLOCK D   OBSERVE                                                           |
+=============================================================================+

   y_d(k,t) = theta_r(k,d) + AWC(k,d) * u^{l(d)}_k(t) + delta_d(k)

      l(0-10 cm)  = surface store u1
      l(10-30 cm) = deep store u2
      l(30-50 cm) = deep store u2

                     y-hat (B, 3, 70, 70)   daily, 32 m
                              |
        +---------------------+--------------------+
        v                     v                    v
   +----------+     +-----------------+    +----------------+
   | L_abs    |     | L_pair          |    | L_lst   (aux)  |
   | Huber at |     | Huber on        |    | Linear(S_k->1) |
   | station  |     | y_i - y_j       |    | vs Landsat     |
   | cell k*  |     | SAME DATE       |    | anomaly        |
   +----------+     +-----------------+    +----------------+
```

---

3. BLOCK A — DESCRIBE

What it does

Produces one 768-number description per location, and one per day. This is `model.py` exactly as it is today: the same sequence, the same six self-attention layers, the same weights. The only addition is `terrain_stem`, a stride-5 convolution over the terrain rasters, added into the 196 spatial tokens with **zero-initialised** weights.

Equations

```
ctx  = Transformer( seq )                   seq in R^(B x 1035 x 768)

S_k  = ctx[:, 12+k, :]                      description of cell k
g    = masked_mean( ctx )                   description of the tile
```

Note what is *not* here: the forcing is **not** read off `ctx`. See §0.5 — that would reopen the memorisation channel §0.4 closes.

Parameters

symbol	shape	source	meaning
`terrain_stem`	~60 k	learned, zero-init	maps (TWI, HAND, slope, curvature) at 32 m into the 160 m token grid
`S_k`	768	output	what cell k looks like, in context of the whole tile
`g`	768	output	what the tile looks like

Logic and foundation

Why terrain enters here, additively, zero-init. Three consequences, each load-bearing. Adding zero changes nothing on day 0, so the forward pass is bit-identical to `best.pt` and §31.8's regression gate passes trivially. Token *k*'s terrain lands on token *k*, whereas appending pooled terrain tokens the `dem_pyr` way would make it tile-level and able only to shift a mean. And `g` receives terrain anyway through attention, because it is computed from the encoder *output*.

Why this block is asked for a description and not a prediction. §27a established that the tokens carry real within-tile spatial heterogeneity at 160 m and that it tracks terrain and land cover — the two positive controls both passed. §27b established that the same tokens do not track wetness. So this block is asked for *what kind of place this is*, which is measured to work, and is not asked for *how wet it is*, which is measured not to.

4. BLOCK B — PARAMETERISE

4.1 B1 — cell parameters

What it does

Converts each location's description into the properties that govern how it holds water.

Equation

```
[ lambda_k , alpha_k , beta_k , p_k , K_k ]  =  g_cell( [ S_{k//5} , fine_k ] )

S_{k//5}   the 768-vector of the 160 m token containing cell k   (gather)
fine_k     measured 32 m channels at cell k: TWI, HAND, slope, curvature,
           S1 sigma0 latest and median, soil texture, LULC
```

Parameters

symbol	name	units	range	forced by	physical meaning
`lambda`	retention	—	(0,1)	sigmoid	fraction of water still present tomorrow under gravity drainage alone
`alpha`	inverse capacity	1/mm	(0, 1/30)	softplus, capped	how much one millimetre of rain raises relative saturation. Small = deep or porous soil
`beta`	ET over capacity	1/mm	(0, 1/20)	softplus, capped	how strongly atmospheric demand draws this cell down. Large = shallow soil, dense vegetation, exposed
`p`	percolation	1/day	(0,1)	sigmoid	fraction of the surface store passed to the deeper store each day
`K`	radar gain	—	(0,1)	sigmoid	how much to trust a Sentinel-1 innovation here

Logic and foundation

Why three dynamic parameters and not the six an obvious design would use. Substituting `u = s/c` into the balance shows that capacity `c` and ET rate `gamma` appear only as the combinations `R/c` and `gamma/c`:

```
s(t+1) = clip( lambda*s + R - E*gamma*s/c , 0, c )

divide by c:

u(t+1) = clip( lambda*u + R/c - E*(gamma/c)*u , 0, 1 )
                          |---|      |-------|
                          alpha         beta
```

`c` never appears alone. There is a one-dimensional family of `(c, gamma)` giving identical predictions, so naming both is naming a parameter the data can never see. Parameterising `alpha = 1/c` and `beta = gamma/c` removes an **exact** degeneracy rather than hoping training resolves it. This reduction came out of the equifinality critique and it makes the model smaller, not larger.

Why every parameter is bounded by its activation. The network cannot create water (`lambda < 1`), cannot have negative capacity, and cannot evaporate what is not there. It chooses inside a box that physics defines. Compare the U-Net, which was free to paint anything and — per §23 — painted a field anti-correlated with the landscape.

Why a shared map rather than per-station parameters. `g_cell` is one MLP applied at all 993 stations. That is the difference between this and classical hydrological calibration, and it is the whole answer to equifinality of predictions — see §10.2.

4.2 B2 — forcing

What it does

Converts each day's weather into two scalars: how much water enters the soil, and how hard the atmosphere pulls.

Equation

```
[ R(t) , E(t) ]  =  softplus( MLP( [ era5_row(t) , g_static ] ) )

g_static = pooled soil / terrain / LULC summary.  Static. Tile-level.
           NO per-date content. NO station-identifying content.
```

Parameters

symbol	name	units	physical meaning
`R(t)`	effective infiltration	mm/day	rainfall that actually enters the soil, after interception, runoff, and freezing
`E(t)`	atmospheric demand	mm/day	what would evaporate if water were unlimited — a potential evapotranspiration

Logic and foundation

Why not raw ERA5 precipitation. ERA5 says 12 mm fell. How much enters depends on whether it was one storm or twelve hours of drizzle, whether there is a canopy intercepting it, whether the ground is already saturated, and whether it is frozen. Conditioning on `g_static` lets the head emit *effective* forcing rather than raw rainfall, which is what the water balance needs.

Why it is shared across all cells. ERA5-Land is 9 km; the tile is 2.24 km. There is no within-tile weather information in the data, and inventing some would be §23's failure repeated in a new place. This is the assumption the whole design rests on: **the weather is shared, and the differences are in the ground.**

Why it must NOT be read off `ctx`. This is §0.5, and it is the single most important constraint in the document. Day-tokens inside `ctx` have attended to the station's own spatial tokens, so reading forcing off them supplies 365 × 2 station-specific, date-specific numbers — a memorisation channel richer than the one that produced the current 21× overfit. The whole benefit of §0.4 depends on this head being blind to per-date station content.

4.3 B3 — readout constants

What it does

Converts relative saturation, which is dimensionless and lives in [0,1], into volumetric soil moisture in m³/m³.

Equation

```
y_d(k,t)  =  theta_r(k,d)  +  AWC(k,d) * u_k(t)  +  delta_d(k)
             |---------- pedotransfer ---------|    |-- learned --|
```

Parameters

symbol	name	units	source	physical meaning
`theta_r`	residual water content	m³/m³	pedotransfer on texture	what a probe reads when the plant-available store is empty — tightly bound water
`AWC`	available water capacity	m³/m³	pedotransfer on texture and bulk density	the swing between wilting point and field capacity
`delta_d`	correction	m³/m³	learned, small, regularised toward zero	sensor calibration, install depth, and the point-versus-pixel support mismatch

Logic and foundation

Why `theta_r` and `AWC` are read from tables, not learned. Two independent reasons. First, they are standard pedotransfer outputs computed from texture and bulk density, and the rasters are already on disk as `soil (21,74,74)` — three depths by seven properties. Second, and decisively for identifiability: `y = A + B*u` means any rescaling of `u` can be absorbed by `B`. The only thing pinning `u`'s scale is the clip at 1, which never engages at a site that never saturates, so at a semi-arid station `B` and `alpha` are genuinely degenerate. Fixing `B` from texture pins the scale unconditionally.

Why the level goes to tables rather than to the network. §27b's ladder measured a tabular stack of soil, terrain, land cover, climate, lat/lon and SMAP at 0.0752 → 0.0576, a skill of 23.4 % on station-mean soil moisture. The trained network's own RMS per-station bias is 0.0618 against the same 0.0752 null — about 18 %. **The network is currently worse at predicting station level than covariates already sitting on disk.** Delegating that job is a win by measurement, not by argument.

Why `delta_d` exists and is kept small. Part of every station's mean is instrument, not landscape — calibration, installation depth, and the mismatch between a point probe and a 32 m cell (§27b.6). No image contains that, and no pedotransfer function does either. It is irreducible, so it gets an explicit, regularised term rather than being allowed to contaminate every spatial parameter in the model.

4.4 B4 — catchment

What it does

Carries the state of the whole upstream catchment as a single number, and distributes it across the tile using the topographic index.

Equations

```
S(t+1)  =  S(t)  +  Q0*exp( -S(t)/m )  -  mean_k( r_k(t) )      deficit, mm
                    |-- baseflow out --|   |-- recharge in --|

D_k(t)  =  S(t)  +  (1/f)*( TWI_mean - TWI_k )                   local, mm
           |moves|         |------ fixed, from §32 ------|

z_k(t)  =  D_k(t) / ( 1000 * n_e )                               depth, m

u*_k(t) =  exp( -z_k(t) / L_k )                                  equilibrium

D_k <= 0  ==>  saturated; u1 = 1 and incoming R becomes runoff
```

Parameters

symbol	name	units	source	physical meaning
`S`	catchment storage deficit	mm	**state**, evolved	how far the whole catchment is from saturation
`m`	recession scale	mm	learned from `g`	how fast baseflow declines as the catchment dries — the catchment's memory
`Q0`	baseflow at zero deficit	mm/day	learned from `g`	discharge when saturated
`f`	transmissivity decay	1/mm	**fixed** at a literature value	how fast soil transmissivity falls with depth; sets the spatial spread of water-table depth
`n_e`	drainable porosity	—	pedotransfer	converts a storage deficit into a depth
`L`	capillary fringe height	m	pedotransfer on texture	how far water rises above the water table. Sand 0.1–0.3, silt 1–2, clay 3–10

Logic and foundation

The problem this solves. A real catchment is tens of km²; the patch is 5 km². You cannot route the catchment — that would need container parameters over 50 km, i.e. the transformer over 50 km. You cannot ignore it — the water arrives regardless. §31.4 measured the consequence directly: **the lowest cell lies on the tile boundary in 25 of 30 sampled tiles (83 %)**, so flow leaves the window and arrives from outside it.

TOPMODEL's answer (Beven & Kirkby 1979): never route it. Carry one catchment-average deficit and distribute it by the topographic index. All the non-local information enters through `TWI_k`, which §32 computes on the full regional flow field and which is therefore correct at catchment scale by construction. This is why §32's 353-region build is a dependency: the TWI it produces is exactly the quantity this block needs, and a tile-local TWI would not do.

Why this makes terrain dynamic — the crux of the design.

```
TWI AS A STATIC INPUT CHANNEL          TWI AS A DEFICIT DISTRIBUTION
-----------------------------          -----------------------------
feeds a network alongside slope,       D_k(t) = S(t) + (1/f)(TWI_mean - TWI_k)
soil, land cover
                                       same fixed TWI, but the water table
can only shift cell k's MEAN           it implies MOVES with the catchment

the terrain effect is identical        the terrain effect is strong when wet
in a drought and a flood               and vanishes when dry
```

`TWI_k` is fixed; `S(t)` moves. So the water table under every cell rises and falls together, but each cell sits at its own offset, set by how much of the catchment drains through it. Same input, computed the same way by §32, completely different functional role — and it is the role TOPMODEL derives it for. §31.5 already states that *"TWI is a water-table-depth index, not a soil moisture index"*; this uses it as one.

Why Grayson's state-dependence falls out rather than being imposed.

```
WET CATCHMENT, S small                DRY CATCHMENT, S large

many cells have D_k <= 0 and          no cell saturates; z_k is deep
saturate; exp(-z/L) is large          everywhere; exp(-z/L) collapses

-> the pattern organises by TWI       -> every cell is an independent 1-D
   "nonlocal control"                    column governed by texture and
                                         vegetation: "local control"
```

Grayson et al. (1997) and the Tarrawarra work (Western, Grayson & Blöschl) established that terrain control of surface moisture is **state-dependent**. §31.5 argued this is what dictates the architecture. Here it is a *consequence* of one state crossing a threshold, spending no free parameters on it. An earlier draft of this design used a learned sigmoid gate on tile wetness; saturation does the same job with two fewer parameters and a derivation behind it.

Why `f` is fixed rather than learned. `m` is identified by the temporal data — 993 daily series constrain a recession rate well. `f` is identified only by the spatial data, which here is the 75 colocated pairs. Classical TOPMODEL's documented equifinality comes from calibrating both against **one outlet hydrograph**, a purely temporal signal, so `f` was never constrained. Here they are constrained by different dimensions of the data — but the spatial dimension is thinly sampled, so `f` is held at a literature value unless §32.8's sufficiency gate demonstrates the pairs can identify it.

---

5. BLOCK C — SIMULATE

Equations

Surface store, 0–10 cm.

```
u1(t+1) = clip[  lambda_k * u1(t)              gravity drainage
               + alpha_k * R(t)                infiltration
               - beta_k * E(t) * u1(t)         evapotranspiration
               - p_k * u1(t)                   percolation to depth
               + Phi1_k(t)                     capillary rise
               ( + kappa_k * [DIV_W u1](k) )   hillslope routing, V6 only
              ,  0 , 1 ]
```

Deep store, 10–50 cm.

```
u2(t+1) = clip[ lambda2_k * u2(t) + p_k * u1(t) - r_k(t) + Phi2_k(t) , 0 , 1 ]

r_k(t) = recharge leaving the deep store to the water table
```

Capillary exchange.

```
Phi_k(t) = omega * max( 0 , u*_k(t) - u_k(t) )
```

Logic and foundation

Why the ET term is multiplied by `u`. A dry soil resists evaporation. This is the standard water-stress limitation, and without it a soil would evaporate at potential rate down to zero, which no soil does.

Why capillary rise is one-sided. The water table can only *supply* water upward. Downward loss is already carried by `lambda` (gravity drainage) and `p` (percolation), so a two-sided relaxation would double-count the loss and over-dry any cell with a deep water table.

Why two stores. The deep store is a low-pass-filtered version of the surface store, so the 10–30 and 30–50 cm bins come out naturally flatter and more dominated by their level than 0–10 cm. That is what the observations do — §21.8 found 10–30 is the *best* depth (ubRMSE 0.0500 vs 0.0539), and §24.12 found identity dominating at depth. Both are consistent with a damped cascade.

Why the clip matters beyond numerics. `u` in [0,1] is what pins the scale of the state. Together with `AWC` fixed from texture (§4.3), it is what stops the readout and the dynamics trading off against each other.

Why `DIV_W` (V6) is optional and off by default. TOPMODEL's `D_k` already handles lateral redistribution at every scale under a quasi-steady water table. `DIV_W` adds only the days-timescale transient in the unsaturated zone that TOPMODEL assumes away. It also reintroduces the boundary problem this design was built to avoid: a 70×70 patch has a wall at its edge, water arrives from outside it, and perimeter cells shed into nothing so their `kappa` is unidentifiable. Running V1–V4 first settles by measurement whether that transient is worth the cost, and whether `DIV_W` and `D_k` are double-counting the same within-patch accumulation.

Sentinel-1 assimilation

```
sigma_hat(k,t) = h( u1_k(t) , roughness_k , vegetation_k )     forward operator
u1_k  <-  clip( u1_k + K_k * ( sigma_obs(k,t) - sigma_hat(k,t) ) , 0 , 1 )
                                  |------- innovation -------|
```

Why the innovation and not the raw value. Absolute backscatter is dominated by surface roughness and vegetation structure, not by water. The *gap between what the state predicted and what the radar saw* is mostly water. §30.3a reaches for the same thing by feeding `(latest, long-term median)` and hoping the network differences them; making it an explicit innovation does that structurally.

Why staleness needs no embedding. Between acquisitions the state coasts forward on ERA5 alone. If there has been no pass for three weeks, the state has not been corrected for three weeks and its error has grown on its own. The staleness embedding at `model.py:505-514` becomes unnecessary on this path.

---

6. BLOCK D — OBSERVE

Equation

```
y_d(k,t) = theta_r(k,d) + AWC(k,d) * u^{l(d)}_k(t) + delta_d(k)

L  =  L_abs  +  w_p * L_pair  +  w_l * L_lst
```

The three losses

`L_abs` — Huber at the station's own cell, per depth. This is today's loss unchanged, with a per-sample cell index replacing the hardcoded `(112,112)` at `model.py:779`.

`L_pair` — Huber on `y_i - y_j` against the observed `y_i - y_j`, for pairs of stations on the same date: 75 colocated pairs across 84 stations, plus every same-ERA5-cell pair.

```
Today nothing in the objective ever compares two locations.

Under an absolute per-pixel loss, the risk-minimising answer at an
unsupervised location is the tile mean -- which is exactly what the model
produces (between-station SD 15-19 % of observed).

Differencing two stations on one date removes weather, season, climate,
the ERA5 cell and the calibration epoch EXACTLY. It is §27b's Scale C
construction applied to supervision instead of to analysis.
```

This is the only term that puts gradient directly on the quantity measured to be broken. It is also the piece §31 lacks: §31 adds the *capacity* for locations to differ but leaves the objective indifferent to whether they do.

`L_lst` — optional auxiliary. `Linear(S_k -> 1)` predicting the Landsat surface-temperature anomaly per token, tile mean removed for that date. Gives Block A dense gradient — roughly 500 independent values per tile against 1, since TIRS is 100 m (§31.6). Four caveats, all from §29 and §31.6: the pattern is **static** (+0.967 month-to-month), so it disciplines the static branch only; the 18.2 % no-retrieval region contains the wettest station, so missingness is correlated with the target; and if the target is a fixed map per station the model may **memorise 993 maps** rather than learn structure — test LST-head skill on validation against training stations in the first smoke run. Decayed and ablatable; nothing downstream reads it.

---

7. COMPLETE PARAMETER INVENTORY

symbol	per	units	source	identified by
`lambda`	cell	—	learned	recession shape, all 2.6 M station-days
`alpha` = 1/c	cell	1/mm	learned	storm response amplitude
`beta` = gamma/c	cell	1/mm	learned	dry-down rate under demand
`p`	cell	1/day	learned	lag between depth bins
`K`	cell	—	learned	error drop on S1 acquisition dates
`R(t)`, `E(t)`	day	mm/day	learned	the whole dataset
`theta_r`	cell, depth	m³/m³	pedotransfer	not fitted
`AWC`	cell, depth	m³/m³	pedotransfer	not fitted
`n_e`, `L`	cell	—, m	pedotransfer	not fitted
`delta_d`	station	m³/m³	learned, regularised	station mean residual
`m`	catchment	mm	learned	**temporal** — 993 daily series
`Q0`	catchment	mm/day	learned	**temporal** — 993 daily series
`f`	catchment	1/mm	**fixed**	**spatial** — only 75 pairs
`TWI`, `HAND`	cell	—, m	§32	not fitted

New network parameters

```
terrain_stem     ~60 k    zero-init
B1 cell head     ~ 8 k
B2 forcing head  ~ 5 k    (includes g_static projection)
B3 delta head    ~ 2 k
B4 catchment     ~ 2 k
S1 operator h    ~10 k
                -------
                ~87 k     against a DELETED four-stage U-Net decoder
```

What is kept, changed, deleted

```
KEPT UNCHANGED
  frozen TerraMind ViT; tokens on disk
  dataset.py sequence assembly (+ terrain added into the spatial tokens)
  the 6-layer fusion transformer and all its weights
  §32's terrain pipeline, exactly as being built

CHANGED
  ctx[:, 12:208]    U-Net bottleneck        ->  five numbers per cell
  masked mean       FiLM on the decoder     ->  catchment params, delta
  ERA5 forcing      averaged into `context` ->  a separate small MLP that is
                                                BLIND to per-date station
                                                content (§0.5)
  the sample        one station-date        ->  one station x 365-day window,
                                                emitting a trajectory

DELETED
  UNetDecoder -- four upsampling stages at 512/256/128/64 + three FiLM'd skips
  the staleness embedding on this path
  (an earlier draft's learned wet/dry gate -- saturation replaces it)
```

---

8. BUILD LADDER

Each rung is a flag with exactly one check. **A rung that fails its check is deleted, not tuned.**

	adds	params	earns its place if
V0	§0.7's CPU experiment	0	bucket ceiling beats 0.0539 **and** parameters are predictable from the landscape. **Gate: everything below is conditional on this.**
V1	per-cell bucket `lambda`, `alpha`, `beta`, one store	~8 k	across-cell SD of `y-hat` rises off zero; `lambda` correlates positively with TWI
V2	`theta_r`, `AWC` from pedotransfer + `delta`	~2 k	r(pred level, obs level) goes from −0.175 to positive
V3	second store, `p`	~1 k	10–30 and 30–50 improve without hurting 0–10
V4	TOPMODEL `S`, `D_k`, capillary coupling	~3 k	r(pred, TWI) **strong when wet, near zero when dry**
V5	Sentinel-1 assimilation	~10 k	error drops on days near an acquisition and not otherwise
V6	near-field `DIV_W` routing	~1 k	anything left after V4 — probably nothing

```
V1 + V2 is a complete, coherent architecture on its own, and it is a
SMALLER change to model.py than deleting the U-Net already requires.

V4 is the intellectually interesting rung -- it is what makes terrain
dynamic and connects to §31.5's theory -- but it should be EARNED by
V1 working, not assumed.
```

---

9. GATES

Before any GPU

```
0.  §0.7's CPU experiment. Bucket ceiling AND parameter predictability.
    This gate can kill the whole design in an afternoon and must run first.

1.  All new heads zero-init  =>  forward bit-identical to best.pt.
    Reuse the §26 provenance gate UNCHANGED: >= 99.5 % of rows
    bit-identical AND max |delta| <= 1 bf16 ULP (0.000977).
    DO NOT re-tighten it.

2.  Mass balance: sum(in) - sum(out) - delta(storage) = 0 to float precision.

3.  Synthetic catchment with analytic TWI: the S recession must match the
    analytic exponential; a plane, a cone and a V-valley must give the TWI
    values §32.5 Tier 1 already specifies (including §32.9.2's correction
    that a cone gives a = r/2 exactly, not one cell).

4.  §32.8's sufficiency gate: regress observed delta-SM on delta-TWI and
    delta-HAND across the 75 pairs under station fixed effects, SPLIT
    WET/DRY. This decides both whether V4 is worth building and whether f
    is identifiable.
```

After training

```
5.  S(t) must be seasonal -- low in spring, high in late summer.
    Flat => the catchment state is inert and V4 is dead weight.

6.  r(prediction, TWI) split by S terciles: STRONG when wet, NEAR ZERO when
    dry. A flat profile is a static offset dressed as terrain and MUST be
    reported as a failure, not as increased variance. This is the same
    statistic as gate 4, so the gate and the architecture test ONE claim.
```

```
 7.  Across-cell SD of y-hat, logged every epoch. Near-zero => collapsed to
     the tile mean -- the failure that looks like success until you plot a map.

 8.  IDENTIFIABILITY. Train 3 seeds; correlate the parameter MAPS, not the
     predictions.
         r > 0.9  => identified. Interpret it, plot it against TWI, publish it.
         r < 0.5  => NOT identified. Predictions may stand; every interpretive
                     claim about that parameter is dropped.
     Plus a flat-direction check: perturb each parameter +/-20 % and measure
     the loss response. A flat direction names the combination to reparameterise
     away, as `c` and `gamma` were in §4.1.

 9.  m, Q0 in physically sane ranges. m = 500 mm means it is fitting noise,
     whatever the RMSE says.

10.  Skill targets, unchanged from §30.9:

         own-centre ubRMSE (0-10)                0.0301  ->  <=
         off-centre ubRMSE                       0.0345  ->  materially closer
         between-station spread, % of observed   15-19 % ->  > 35 %
         r(pred level, obs level), CR200-18      -0.175  ->  > 0

11.  PHYSICALITY, not just variance. norm-std must DROP at SCAN Crossroads --
     the flat control currently receiving the most painted variation (0.52).
     A run that increases spread while r(anomaly, terrain) stays near zero has
     made the map more variable, not more correct, and must not be reported as
     success. §23 is explicit on this.

12.  Report 14x14 and 70x70 SEPARATELY. If 70x70 helps only via the static
     branch, that is a resolution-limited offset, not resolved dynamics.
```

---

10. KNOWN WEAKNESSES, STATED BEFORE BUILDING

10.1 §27b's ceiling may be real

If the imagery genuinely carries no information about how a place holds water, B1 outputs near-constants and every cell in the tile predicts one curve. **No architecture manufactures information.** What this design does is fail *legibly* — a flat `lambda` map says so directly, where a decoder's wrong map looked plausible for two years.

§0.7 is designed to find this out before the allocation is spent.

10.2 Equifinality of parameters is NOT cured

This distinction matters and should not be blurred:

```
EQUIFINALITY OF PREDICTIONS     largely cured by regionalisation. Classical
                                calibration fits a free parameter vector to
                                ONE catchment's hydrograph, which is
                                underdetermined. Here theta_k = MLP(landscape)
                                is ONE map that must work at 993 stations, so
                                a compensating error must be a CONSISTENT
                                function of slope, TWI, texture and land cover
                                or it breaks somewhere else.

EQUIFINALITY OF PARAMETERS      NOT cured. The MLP can settle anywhere on a
                                compensating manifold, and which point it
                                picks is arbitrary.

WHAT IS THEREFORE AT RISK       gates 6 and 8 -- the interpretive story --
                                NOT the soil moisture predictions.
```

The failure mode is: **the model works, and the story about why is unfalsifiable.** Gate 8 decides whether the story may be told. That is a real scientific cost and it should not be papered over.

10.3 The near-field boundary problem is real, and V6 reintroduces it

A 2.24 km patch does not contain a catchment. §31.4 measured the lowest cell on the tile boundary in 25 of 30 tiles. V1–V5 avoid this entirely by never routing water across the patch — TOPMODEL carries the catchment as one number instead. V6 puts it back, and with it: a wall at the patch edge, missing external inflow, and perimeter cells whose `kappa` is unidentifiable because they shed into nothing. That is why V6 is last and off by default.

10.4 TOPMODEL's quasi-steady assumption

TOPMODEL assumes the water table equilibrates instantly with the catchment deficit. In flashy catchments that is wrong. V6 is the patch, and V6 has §10.3's problem. There is no clean answer to this within the design; it is a known limit of the approach, and gate 5 will show it as a poorly-behaved `S(t)`.

10.5 Getting the physics wrong is a failure mode a black box does not have

A misspecified capillary term is a bug that training cannot route around. §31's per-location transformer cannot be wrong about physics because it claims none. That is the trade being made here, deliberately: **this model states its mechanism, so the mechanism can be wrong in a detectable way.**

10.6 The optimisation regime is untested

~56 steps per epoch at batch 128 with 1,095 labels per sample is a different regime from the current one. Rich gradients, few steps. Nothing in this codebase's history speaks to it. Mitigation is random window-start augmentation, which is free, but the risk is real and belongs in the smoke run.

10.7 It will not produce structure below 32 m

And must not be presented as doing so. §23's verdict — *"do not present this model as producing 10 m soil-moisture maps"* — applies to 32 m here just as firmly. Blockiness at 5×5 cells within a token is a **diagnostic**, not a defect: it is the map saying the fine channels are contributing nothing. Bilinear smoothing hides exactly that failure behind a plausible ramp, which is how the current map looked reasonable while being anti-correlated with the landscape. Do not smooth it. Report it.

10.8 `f` rests on 75 pairs

Held fixed at a literature value unless gate 4 says otherwise.

APPENDIX A — the loop, with numbers

Two cells. Same weather. Illustrative parameters, TOPMODEL coupling omitted for clarity; the lateral term shown here is V6's `DIV_W`, used because it makes the mechanism visible in three lines of arithmetic.

```
                    capacity   retention   ET     shedding
---------------------------------------------------------------
HOLLOW               100 mm      0.95      0.8      0.02      holds
RIDGE                 60 mm      0.80      1.0      0.15      sheds

The hollow has three ridge-like cells flowing into it. Gate at 0.6.
```

Day 1 — 20 mm rain, 3 mm demand. Start: hollow 40 mm, ridge 20 mm.

```
RIDGE      s = 20,  fullness 0.333       HOLLOW     s = 40,  fullness 0.400
  rain          + 20.00                    rain          + 20.00
  evaporation   -  1.00                    evaporation   -  0.96
  drains down   -  4.00                    drains down   -  2.00
  sheds sideways-  1.80                    receives      +  4.92
                --------                                 --------
  s(day 2)      =  33.20  fullness 0.553   s(day 2)      =  61.96  f 0.620
```

Day 2 — no rain, 5 mm demand.

```
RIDGE      s = 33.20                      HOLLOW     s = 61.96
  evaporation   -  2.77                     evaporation   -  2.48
  drains down   -  6.64                     drains down   -  3.10
  sheds sideways-  2.99                     receives      +  8.22
                --------                                  --------
  s(day 3)      =  20.80  fullness 0.347    s(day 3)      =  64.60  f 0.646
                         FELL 37 %                                 ROSE
```

The hollow keeps rising the day after rain while the ridge collapses, because water is still arriving from upslope. That lag is real hydrology and it is exactly what one shared weather vector structurally cannot produce.

Through Block D:

```
             LEVEL    RANGE    fullness      soil moisture
---------------------------------------------------------
HOLLOW       0.15  +  0.20  x   0.646    =      0.279
RIDGE        0.12  +  0.16  x   0.347    =      0.176
                                            ---------
                                 spread  =      0.103
```

Against an observed within-tile spread of ~0.108 at CR200-18, where the current model produces ~0.021.

APPENDIX B — comparison with §31

	today	**§31 per-location processor**	**this**
frozen ViT	yes	yes	yes
fusion transformer	1×	1×	1×
per-location model	**none**	transformer, run 196× at inference	3–5 scalars + a recurrence
forcing	inside the fusion sequence, averaged into one vector	`W`, cross-attended per location	a small MLP, blind to station identity
decoder	U-Net, 4 stages	deleted	deleted
new parameters	—	~2 M	~87 k
labels per forward pass	1	1	up to 1,095
terrain's role	—	static input channel	distribution function of a moving deficit
what stops memorisation	nothing (21× overfit)	nothing structural	5 bounded scalars
interpretable	no	no	yes — and therefore falsifiable
can be wrong about physics	no	no	**yes**

§31 and this design agree on the diagnosis and on deleting the U-Net. They disagree on what the per-location object should be: a 768-dimensional transformer output, or a handful of bounded physical scalars. Under one label per tile across 993 stations, the second is a well-posed estimation problem and the first is not — that is the whole argument.

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